NeuroVision-Pi4 | Complete Recovery & Rebuild Manual

Consolidated reference edition — 23 October 2026. Single comprehensive document. This manual preserves the earlier source-inspected recovery procedures and incorporates verified live-system inventory and exact installed Python package versions. English-language developer documentation.

Evidence key: VERIFIED LIVE = observed on the running Pi; VERIFIED SOURCE = inspected in repository code; PROCEDURE = proposed restoration steps not yet end-to-end tested; MISSING BACKUP = information not available in the conversation and never invented.

NeuroVision-Pi4 | Full Recovery & Rebuild Manual Revision: 22 October 2026 | Source: live GitHub repository drpeker/NeuroVision-Pi4, default branch neurovision-raspi-4. This document is written in English for reproducible global development.

Scope: Raspberry Pi 4 (neurobrain), XiaoZhi voice assistant, Logitech C270 camera, SSD MobileNet v2 vision service, MJPEG, and MCP get_visible_objects. The separate Raspberry Pi Zero 2 W (neuroeye) and offline llama.cpp brain are NOT automatically restored by this repository.

1. Recovery assurance and limitations

VERIFIED: code paths, camera configuration, startup script, MCP registration, and declared project dependencies were inspected directly from GitHub. NOT VERIFIED: actual machine package versions, hidden credentials, audio device indexes, exact Python venv lock state, OS image, network settings, model binary integrity, and current working state. Do not mistake repository contents for a complete disk image.

IMPORTANT: GitHub read access succeeded, but the attempted README update returned HTTP 403 (integration cannot write). This PDF does not modify the repository.

2. Known machine and architecture

Host: neurobrain; user: drpeker; board: Raspberry Pi 4 Model B, 4 GB RAM; OS reported Debian 13 Trixie, ARM64; USB Logitech C270 camera/microphone; analog audio and external amplifier. Application root: /home/drpeker/projects/neurovision. Startup entry: ~/.local/bin/neuro-vision.

Pipeline: C270 -> OpenCV V4L2 -> RGB/PIL -> LiteRT SSD MobileNet v2 -> /detections JSON and /stream MJPEG -> MCP get_visible_objects -> XiaoZhi voice AI -> Turkish spoken output. The voice path depends on separately provisioned activation credentials and audio configuration.

3. Before wiping anything: capture a real recovery snapshot

Run these on the EXISTING Pi before reinstalling. They collect versions and configuration evidence without exposing secrets in public repositories. Review the resulting archive locally.

```
mkdir -p ~/neurovision-recovery
cd ~/neurovision-recovery
date -Is > timestamp.txt
```

```
uname -a > kernel.txt
cat /etc/os-release > os-release.txt
python3 --version > python.txt 2>&1
dpkg-query -W > dpkg-packages.txt
python3 -m pip freeze > system-pip-freeze.txt 2>&1
lsusb > usb.txt
v4l2-ctl --list-devices > video-devices.txt 2>&1
arecord -l > audio-inputs.txt 2>&1
aplay -l > audio-outputs.txt 2>&1
ip -brief address > ip-addresses.txt
systemctl --type=service --state=running > services.txt
df -h > disks.txt
lsblk -f > partitions.txt
cd ~/projects/neurovision
git remote -v > ~/neurovision-recovery/git-remotes.txt
git status --short > ~/neurovision-recovery/git-status.txt
git rev-parse HEAD > ~/neurovision-recovery/git-commit.txt
.venv/bin/python -m pip freeze > ~/neurovision-recovery/venv-freeze.txt 2>&1
readlink -f ~/.local/bin/neuro-vision > ~/neurovision-recovery/launcher.txt
```

Also securely back up user-specific configuration, activation data, device registration, .env files, audio preferences, downloaded model binaries, and local changes NOT committed to Git. Do not upload credentials

or tokens to a public repository. Prefer an offline encrypted backup or a full SD/SSD image.

4. Rebuild operating system and connectivity

Flash a compatible Raspberry Pi OS/Debian ARM64 image using Raspberry Pi Imager. The original system was reported as Debian 13 Trixie; if using another image, dependency behavior may differ. Set hostname neurobrain, user drpeker, SSH access, locale, timezone, and Wi-Fi/Ethernet. Do not assume password or IP address.

```
hostnamectl
cat /etc/os-release
uname -m
sudo systemctl enable --now ssh
ssh drpeker@neurobrain.local
sudo apt update
sudo apt install -y git curl ca-certificates build-essential \
  python3 python3-venv python3-pip python3-dev \
  ffmpeg v4l-utils alsa-utils libportaudio2 portaudio19-dev \
  libsndfile1 libopus0 libgl1 libglib2.0-0
```

The apt package list is a reasonable bootstrap, not an exact captured manifest. Compare with dpkg-packages.txt from the old installation; repository dependencies may need additional libraries.

5. Restore repository: correct branch, not upstream XiaoZhi

```
mkdir -p ~/projects
cd ~/projects
git clone --branch neurovision-raspi-4 \
  https://github.com/drpeker/NeuroVision-Pi4.git neurovision
cd ~/projects/neurovision
git branch --show-current
git log -1 --oneline
git status --short
```

Do NOT replace this fork with a fresh clone of huangjunsen0406/py-xiaozhi: the fork contains the NeuroVision-specific MCP registration and vision service. Preserve the exact commit hash from the original machine when available.

6. Python environment: reproducible procedure

The upstream README-XIAOZHI.md states Python 3.10–3.12; the project pyproject.toml specifies >=3.10. For a conservative recovery use Python 3.12 if available, not an untested Python 3.13 environment. The repository recommends uv sync; use a lockfile if the repository contains one.

```
cd ~/projects/neurovision
python3 --version
command -v uv || echo "Install uv from its official instructions"
uv python install 3.12
uv sync --python 3.12
uv run python -c "import cv2, PIL, numpy; print('core imports OK')"
```

CRITICAL: the vision code imports ai_edge_litert.interpreter.Interpreter. The inspected pyproject.toml and requirements.txt do NOT explicitly declare ai-edge-litert. Install an ARM64-compatible LiteRT package only after verifying available wheel support and record its exact version. Do not blindly use an incompatible x86 wheel.

```
cd ~/projects/neurovision
.venv/bin/python -c \
  "from ai_edge_litert.interpreter import Interpreter; print('LiteRT OK')"
.venv/bin/python -c "import cv2; print(cv2.__version__)"
```

If uv is unavailable, create a Python 3.12 virtual environment and install requirements after checking platform constraints. The pip and uv resolution paths are not necessarily identical.

7. Vision model files: mandatory

The vision service uses EXACT relative paths vision/models/ssd_mobilenet_v2.tflite and vision/models/coco_labels.txt. Both must exist; the model binary cannot be regenerated from labels alone. Copy any missing .tflite binary from your verified backup or a validated original release.

```
cd ~/projects/neurovision
ls -lh vision/models/ssd_mobilenet_v2.tflite \
       vision/models/coco_labels.txt
sha256sum vision/models/ssd_mobilenet_v2.tflite
head -10 vision/models/coco_labels.txt
```

Labels are zero-based COCO-style mappings in the inspected file. Preserve matching class ordering and output tensor interpretation. Do not substitute a different TFLite model without verifying its input shape, tensor ordering, and labels.

8. Camera capture and model inference: exact code settings

Source: vision/neurovision_live.py. Camera cv2.VideoCapture(0, cv2.CAP_V4L2); FOURCC MJPG; requested width 640; height 480; FPS 30. Model threshold 0.40; LiteRT interpreter num_threads=2. Captured BGR frames are converted to RGB; PIL resizes to the model input dimensions; floating-point inputs are normalized with (x - 127.5) / 127.5. Output tensors are assumed boxes, classes, scores, count, in that order.

JPEG output uses quality=70. HTTP server binds 0.0.0.0:8080 and serves /, /stream, /detections. MJPEG sends a frame approximately every 0.2 s to each client, which is NOT equivalent to guaranteed 30 FPS inference or stream delivery.

```
sudo apt install -y v4l-utils
v4l2-ctl --list-devices
v4l2-ctl -d /dev/video0 --list-formats-ext
ls -l /dev/video*
groups
cd ~/projects/neurovision
.venv/bin/python vision/neurovision_live.py
```

If the camera is not /dev/video0, inspect the V4L2 device list and change the index only after confirming the correct node. USB video device numbering can change between boots. If access is denied, inspect permissions and group membership (often video).

9. Camera endpoint validation

```
curl -fsS http://127.0.0.1:8080/detections
curl -I http://127.0.0.1:8080/
ss -ltnp | grep ':8080'
# Browser on local network:
# http://neurobrain.local:8080/
# http://neurobrain.local:8080/stream
# http://neurobrain.local:8080/detections
```

The detection endpoint can return [] when no object is detected; this alone is not an error. /stream is multipart MJPEG, not a single JPEG. The server currently has no authentication and binds all interfaces; use only on a trusted LAN or add access controls before exposure.

10. MCP integration: exact source and wiring

File src/mcp/tools/neurovision/register.py registers an MCP tool named get_visible_objects with an empty PropertyList. It calls http://127.0.0.1:8080/detections using urllib.request.urlopen with timeout=3 seconds and returns JSON containing source, objects, and a Turkish description instruction. If unavailable, it returns an error JSON payload.

File src/mcp/mcp_server.py, method add_common_tools, imports register_neurovision_tools from src.mcp.tools.neurovision and invokes register_neurovision_tools(self.add_tool). The src/mcp/tools/neurovision/__init__.py re-exports that function. This explicit registration is a CRITICAL fork-specific modification.

```
grep -n 'register_neurovision_tools' \
  src/mcp/mcp_server.py src/mcp/tools/neurovision/*.py
grep -n 'get_visible_objects\|127.0.0.1:8080' \
  src/mcp/tools/neurovision/register.py
grep -R -n 'MCP_TOOLS' . --exclude-dir=.git | head
```

If MCP_TOOLS.DISABLED contains get_visible_objects, it will not be advertised. The server uses JSON-RPC 2.0, supports initialize, tools/list, tools/call, and returns paginated tool lists. Do not assume an MCP tool is working merely because the camera webpage opens: test an actual tool call through the XiaoZhi protocol.

11. Startup script and exact runtime order

Source: scripts/neuro-vision. The script changes to /home/drpeker/projects/neurovision, selects $PWD/.venv/bin/python, checks /detections on port 8080, launches the vision service with setsid, waits up to 40 x 0.5 s (roughly 20 seconds plus HTTP overhead), then starts main.py --mode cli. Ctrl+C should trigger cleanup of the camera process group.

```
cd ~/projects/neurovision
chmod +x scripts/neuro-vision
mkdir -p ~/.local/bin
ln -sfn "$HOME/projects/neurovision/scripts/neuro-vision" \
  "$HOME/.local/bin/neuro-vision"
~/.local/bin/neuro-vision
```

If port 8080 is occupied, the script deliberately refuses to launch. Do not kill unrelated processes without checking ownership. If .venv/bin/python does not exist, rebuild the virtual environment first. Do not run two independent copies of the vision server.

12. XiaoZhi activation, audio and network

The upstream application uses online WebSocket/MQTT services and device authentication. The repository does not reveal the user-specific activation credentials or currently selected server configuration. Restore these from a secure backup or re-register the device through the service. Do not fabricate activation tokens or hardcode secrets in README files.

```
arecord -l
aplay -l
python3 -c "import sounddevice as sd; print(sd.query_devices())"
cd ~/projects/neurovision
.venv/bin/python main.py --mode cli
```

Verify that the selected microphone is actually the Logitech C270 or intended USB microphone, and that playback uses the desired analog audio device/amplifier. Test capture and playback separately before debugging network AI responses. Check ALSA/PulseAudio/PipeWire device routing and volume; these

were not independently captured.

13. Failure diagnostics: symptoms and checks

Camera unavailable: check lsusb, v4l2-ctl, /dev/video*, permissions, camera index, power and USB cable. LiteRT import error: verify architecture, wheel and venv interpreter. Empty detections: verify actual frames, lighting, 0.40 threshold, matching labels and tensor ordering. Wrong colors: ensure BGR-to-RGB conversion remains intact.

MCP tool missing: check register_neurovision_tools in mcp_server.py, __init__.py export, and disabled-tool configuration. MCP errors: curl localhost:8080/detections from the SAME Pi, then inspect timeout or server logs. No XiaoZhi voice: test audio I/O, service registration, authentication, network, and server logs. Port collision: ss -ltnp | grep 8080.

```
ps aux | grep -E 'neurovision_live|main.py' | grep -v grep
ss -ltnp | grep 8080
free -h
dmesg -T | tail -60
journalctl -b --no-pager -n 100
git diff --stat
git status --short
```

14. Backups, security and long-term reproducibility

Create a tagged known-good commit after testing. Save SHA256 checksums for model binaries, a dependency lock, hardware inventory, sanitized configuration template, and recovery test results. Back up actual credentials separately and encrypted. Use git status before every destructive command. A Git clone restores tracked source code, NOT your device registration, .venv, model files omitted by Git, OS services, or local settings.

```
cd ~/projects/neurovision
git status --short
git rev-parse HEAD
sha256sum vision/models/*.tflite > ~/neurovision-recovery/models.sha256
tar -czf ~/neurovision-source-snapshot.tar.gz \
  --exclude=.git --exclude=.venv -C ~/projects neurovision
```

A full block-device image is the only reliable way to preserve unknown system-level customizations exactly. Verify backups by restoring onto a spare SD/SSD before erasing the working system.

15. Final acceptance checklist

[ ] Host neurobrain resolves over LAN; SSH login works for drpeker.

[ ] Correct fork branch and commit checked out; no unintended uncommitted changes.

[ ] Python venv exists and imports cv2, PIL, numpy, and LiteRT.

[ ] SSD MobileNet v2 binary and coco_labels.txt exist and are compatible.

[ ] Logitech C270 detected; frames read at a verified negotiated format.

[ ] HTTP /, /stream and /detections respond locally and from intended LAN client.

[ ] get_visible_objects is advertised and returns detection JSON through MCP.

[ ] Microphone capture, speaker output, and XiaoZhi activation independently pass.

[ ] Launcher starts both processes and Ctrl+C terminates them cleanly.

[ ] Offline encrypted backup and validated disk image exist.

16. Source inventory and important distinction

GitHub repository: https://github.com/drpeker/NeuroVision-Pi4 ; branch neurovision-raspi-4. Files directly inspected: README.md; README-XIAOZHI.md; scripts/neuro-vision; vision/neurovision_live.py; vision/models/coco_labels.txt; src/mcp/tools/neurovision/register.py; src/mcp/tools/neurovision/__init__.py; src/mcp/mcp_server.py; pyproject.toml; requirements.txt.

Other remembered projects are distinct: Raspberry Pi Zero 2 W neuroeye has its own NeuroEye repository and CSI OV5647 camera; the earlier offline Raspberry Pi 4 Qwen/Whisper/Piper stack has different startup and model paths. Neither should be silently merged into this XiaoZhi recovery procedure. Their complete rebuild requires their respective source and current device inventories.

17. VERIFIED LIVE baseline — authoritative for this installation

VERIFIED LIVE (2026-10-23 00:11 +03:00): hostname neurobrain, user drpeker; Debian GNU/Linux 13.7 (trixie); Linux 6.18.50+rpt-rpi-v8, aarch64; Python 3.13.5; Raspberry Pi 4, approximately 3.7 GiB usable RAM; swap 2.0 GiB; /dev/mmcblk0p2 root filesystem 58 GiB, 19 GiB used, 37 GiB free.

VERIFIED LIVE: Git repository /home/drpeker/projects/neurovision; HEAD da9539b9cb542cae6eb44ea823afde00c97be997; git status --short produced no entries at inventory time. Preserve the correct fork and commit.

VERIFIED LIVE: USB Logitech C270, vendor:product 046d:0825. V4L2 device /dev/video0 (also /dev/video1); negotiated 640x480, MJPG, 30/1 FPS, sRGB. This is the V4L2 configuration, NOT a measured AI inference rate.

VERIFIED LIVE: ALSA capture C270 HD WEBCAM card 3 device 0; analog playback bcm2835 Headphones card 0 device 0; HDMI playback cards 1 and 2. sounddevice enumerated only playback devices 0,1,2 and default [-1,2] while the application was running. Do not infer a microphone failure or alter a working audio setup. ALSA card numbers and PortAudio indices are not stable identifiers.

VERIFIED LIVE: virtual environment is uv-managed; .venv/bin/python -m pip returned No module named pip. This is not a failure. uv pip list successfully enumerated installed packages; uv pip freeze was subsequently executed successfully to write ~/neurovision-recovery/python-packages.lock.txt. The contents of that file have not been independently read back.

18. Complete observed Python package inventory

VERIFIED LIVE: The following names and versions were copied from the successful uv pip list output, not inferred from pyproject.toml. The on-device python-packages.lock.txt is the machine-readable primary snapshot. Editable project installation is noted separately.

```
ai-edge-litert 2.3.0
aiohappyeyeballs 2.6.1
aiohttp 3.13.2
aiosignal 1.4.0
altgraph 0.17.5
annotated-types 0.7.0
anyio 4.12.0
attrs 25.4.0
backports-strenum 1.3.1
certifi 2025.11.12
cffi 2.0.0
charset-normalizer 3.4.4
colorzero 2.0
cryptography 46.0.3
distro 1.9.0
evdev 1.9.2
flatbuffers 25.12.19
frozenlist 1.8.0
gpiozero 2.0.1
h11 0.16.0
httpcore 1.0.9
httpx 0.28.1
idna 3.11
iniconfig 2.3.0
jiter 0.12.0
lgpio 0.2.2.0
lunar-python 1.4.8
ml-dtypes 0.6.0
multidict 6.7.0
mutagen 1.47.0
numpy 2.4.0
openai 2.14.0
opencv-python-headless 4.11.0.86
opuslib 3.0.1
packaging 25.0
paho-mqtt 2.1.0
pendulum 3.1.0
pillow 12.1.0
platformdirs 4.5.1
pluggy 1.6.0
propcache 0.4.1
protobuf 7.36.2
psutil 7.2.1
py-machineid 1.0.0
pycparser 2.23
pydantic 2.12.5
pydantic-core 2.41.5
pygments 2.19.2
pyinstaller 6.17.0
pyinstaller-hooks-contrib 2025.11
pynput 1.8.1
```

```
pyperclip 1.11.0
pypinyin 0.55.0
pytest 9.0.2
pytest-asyncio 1.3.0
python-dateutil 2.9.0.post0
python-xlib 0.33
requests 2.32.5
ruff 0.14.10
setuptools 80.9.0
sherpa-onnx 1.13.0
sherpa-onnx-core 1.13.0
six 1.17.0
sniffio 1.3.1
sounddevice 0.5.3
soxr 1.0.0
tqdm 4.67.1
typing-extensions 4.15.0
typing-inspection 0.4.2
tzdata 2025.3
urllib3 2.6.2
websockets 15.0.1
yarl 1.22.0
```

Editable package: py-xiaozhi 1.0.0 -> /home/drpeker/projects/neurovision. The package listing is a snapshot of installed distributions; exact rebuild also depends on Python ABI, ARM64 wheels, apt libraries, model binary, configuration and service credentials.

19. Corrected Python restoration procedure — preserve 3.13.5

IMPORTANT CORRECTION: The earlier source-inspected manual recommended Python 3.12 based on upstream README compatibility. The ACTUAL WORKING Pi uses Python 3.13.5, so matching 3.13.5 is the preferred first restoration attempt. Do not downgrade a working installation merely to satisfy upstream documentation. Use uv and the saved freeze snapshot; do not install pip into the original environment just for inventory.

```
cd ~/projects/neurovision
uv --version
uv python install 3.13.5
# Before replacing any existing .venv, keep a copy of it.
# On a clean rebuild only:
uv venv --python 3.13.5 .venv
uv pip install --python .venv/bin/python -r ~/neurovision-recovery/python-packages.lock.txt
uv pip install --python .venv/bin/python -e .
.venv/bin/python -c "import cv2,numpy,PIL,sounddevice; from ai_edge_litert.interpreter import Interpreter; pri
```

PROCEDURE, NOT TESTED END-TO-END: The freeze file may contain an editable local path. On a clean Pi, install the local project with -e . after restoring the third-party packages; if uv pip install -r fails on an editable entry, remove only that entry from a COPY of the lock and install -e . separately. Prefer an existing uv.lock if present, but compare resolved versions against this verified inventory. The exact freeze file was created on the Pi and must be preserved off-device; it is not embedded here because its contents were not supplied.

20. Restore order, without damaging the working system

1. Before any wipe, back up the entire boot medium as an image if feasible. Keep a second copy of the repository, the model .tflite, any untracked files, ~/neurovision-recovery/python-packages.lock.txt, and private application/device credentials in encrypted offline storage.

2. On a fresh spare SD card, install Debian 13 ARM64 or the closest validated Raspberry Pi image; restore hostname/user and network. Install OS libraries as listed in section 4. Record any deviations.

3. Clone drpeker/NeuroVision-Pi4 branch neurovision-raspi-4, then check out commit da9539b9cb542cae6eb44ea823afde00c97be997 if it is present. Never overwrite the live working tree.

4. Restore Python 3.13.5 and uv-managed venv; use the off-device saved freeze file. Verify LiteRT and OpenCV imports before proceeding.

5. Restore the exact SSD MobileNet v2 TFLite binary and matching labels, compare saved checksums. Camera uses MJPG 640x480 at 30 FPS; validate /dev/video0 on the new installation.

6. Start vision service and verify /detections and /stream. Confirm port 8080 is free and that the service is LAN-only unless authenticated.

7. Confirm MCP registration in src/mcp/mcp_server.py and get_visible_objects in src/mcp/tools/neurovision/register.py; test the MCP call, not just the browser endpoint.

8. Restore XiaoZhi service credentials and device activation privately; confirm microphone and analog output with the real application. Avoid hardcoding ALSA numeric card indexes.

9. Install or verify ~/.local/bin/neuro-vision symlink to scripts/neuro-vision, then run the combined launcher and confirm clean exit.

10. Compare the spare-card restore against the live system and save the final known-good disk image. Do not erase the original until acceptance tests pass.

21. Project development history and architectural decisions

The repository began from the upstream py-xiaozhi voice assistant. The fork adds a Raspberry Pi 4 vision pipeline, rather than replacing XiaoZhi voice transport. The goal is camera-based perception exposed to the assistant as an MCP tool.

Vision architecture: USB Logitech C270 -> OpenCV V4L2 MJPG -> BGR-to-RGB -> PIL/model input -> LiteRT SSD MobileNet v2 COCO object detection -> in-process JSON state and JPEG frames -> HTTP /detections and /stream -> MCP get_visible_objects -> voice assistant response. This is object detection, not a general vision-language model.

Design choices: 640x480 MJPG reduces USB capture overhead; 30 FPS is camera configuration, while MJPEG server emits frames approximately every 0.2 seconds; LiteRT uses two inference threads and score threshold 0.40. Avoid confusing camera FPS, stream FPS, and model inference FPS.

The fork-specific registration is essential: register_neurovision_tools(self.add_tool) in src/mcp/mcp_server.py. If this call disappears during an upstream merge, XiaoZhi can still run but will no longer have the get_visible_objects tool.

The launcher scripts/neuro-vision coordinates the two processes and avoids duplicate servers on port 8080. Preserve its executable bit, absolute working directory, and cleanup behavior. The system has no demonstrated always-on systemd service in the inspected evidence.

Other development efforts are separate: Raspberry Pi Zero 2 W neuroeye / NeuroEye repository uses an OV5647 CSI camera; the offline NeuroBrain Qwen/Whisper/Piper/llama.cpp work is not recreated by this XiaoZhi fork. Do not merge their dependencies or service instructions without a separate inventory.

22. Backup artifacts still required for complete disaster recovery

MISSING BACKUP: contents of the actual python-packages.lock.txt (created successfully on the Pi, but not transferred into this document); retain this file outside the Pi.

MISSING BACKUP: full apt/dpkg package manifest, exact uv.lock status, model binary SHA256 and a separately copied .tflite binary, device-specific secrets and activation tokens, private config and environment files, any non-Git scripts or service units, ALSA/PipeWire custom configuration, SSH/network settings and an actual disk image.

GitHub repository source alone cannot reconstruct secrets, untracked data, hardware state, and model binaries excluded from Git. Keep secrets in an encrypted backup, never a public repository. A successful restoration on spare media is the definitive test.

23. Quick acceptance matrix

```
hostnamectl
python3 --version
git -C ~/projects/neurovision rev-parse HEAD
uv pip list --python ~/projects/neurovision/.venv/bin/python
lsusb
v4l2-ctl -d /dev/video0 --get-fmt-video --get-parm
ls -lh ~/projects/neurovision/vision/models/ssd_mobilenet_v2.tflite
curl -fsS http://127.0.0.1:8080/detections
ss -ltnp | grep 8080
arecord -l
aplay -l
```

Use these commands for inspection on a restored installation; the camera service must be running for the HTTP check. Final success requires a real spoken XiaoZhi interaction that invokes get_visible_objects, correct visual objects, working audio output and safe shutdown.